Joseph Mesches

jmesches@gmail.com | linkedin.com/in/joe-mesches | github.com/jkmesches

Electrical Engineering student at CSU and critical-care flight paramedic focused on PCB design, embedded systems, and hardware bring-up. Designs and deploys custom mixed-signal hardware, from schematic and layout through firmware and field deployment.

Education

Bachelor of Science, Electrical Engineering Expected May 2027
Colorado State University Major GPA: 3.7 | Cumulative: 3.4 | Dean's List | HKN
Coursework: Digital System Design, Microcontrollers & IoT, Electronics, Electromagnetic Fields, Digital Logic, Circuit Theory

Associate of Applied Science, Paramedicine December 2019
Arapahoe Community College

Technical Skills

Design & EDA: Altium Designer, KiCad, LTSpice, Multisim, Cadence Virtuoso, Quartus Prime
Embedded: ESP32, ARM, Arduino, embedded C/C++, firmware, battery management, sensor integration
Programming: C/C++, Python (NumPy, SciPy, Pandas), Verilog, ARM Assembly, MATLAB, Bash
Software: Git, Linux systems administration, Docker, Proxmox, Grafana
Lab & Protocols: oscilloscopes, logic analyzers, VNA, spectrum analyzers, FPGAs, soldering, microsoldering; SPI, I2C, UART, CAN, JTAG

Experience

Student Research Assistant, Radar Systems Aug 2026 – Present
Prof. Chandrasekar's Radar Lab, Colorado State University
– Developing AQPI Sentinel, a radar-product monitoring interface, and integrating it into the lab's backend systems.

Electrical Engineering Intern June 2026 – Present
The RMH Group, Inc. (MEP)
– Reverse-engineered an automatic racking robot and conveyor system into AutoCAD drawings; contributed to control-system projects improving plant efficiency and maintained as-built packages (AutoCAD 2024, Adobe Acrobat Pro); invited to continue through the academic year.

Projects

OpenOrbit Radar Payload Team (Senior Design) Aug 2026 – May 2027
Project Manager, Colorado State University
– Lead the team building a drone-based synthetic aperture radar (SAR) imaging system; manage schedule, budget, funding, and sponsor relations while contributing to the engineering.

Rodent Activity Monitor Fall 2025
CU Anschutz Neurotechnology Center
– Designed a custom ESP32 PCB (PIR, CO2, temperature, humidity sensors plus battery management); 5 deployed units with 96-hour to 14-day battery life on configurable sampling.
– Built a full-stack pipeline (Docker, database, web UI, Grafana) now in daily use by lab non-engineers.

Low-Frequency Spectrum Analyzer Spring 2024
Colorado State University
– Led a 3-person team designing a battery-powered ESP32 handheld with TFT display; implemented FFT and windowing in C++ for 1–2 Hz resolution.

Leadership

IEEE Student Branch: President (2025–2026), Vice President (2026–Present) 2025 – Present
Colorado State University
– Lead branch operations, outreach, and communication between students, faculty, and industry partners.

EMS Field Training Officer, Instructor & QA Reviewer 2015 – 2022
Multiple Colorado EMS Agencies
– Led field-training teams of up to 30 and trained 200+ paramedics; served on quality-assurance and education committees authoring SOPs.

Clinical Experience

Critical Care / Flight Paramedic 2017 – 2022
UCHealth LifeLine, Air Methods Corporation, Northglenn Ambulance
– Depended on and troubleshot life-critical medical devices (monitor/defibrillators, ventilators, infusion and balloon pumps) delivering critical care across scene, ED, and inter-ICU transport.

Publications, Honors & Certifications

Publication: "Looking Past Dementia Reveals Hidden Life Threats," *Journal of Emergency Medical Services* (2018)
Certifications: Certified Flight Paramedic (FP-C, ICAPP), Critical Care Endorsement (P-CC), Nationally Registered Paramedic
Honors: Walker-Weinshenker Family Scholar (2024–2026), Dean's List, IEEE-HKN